

A RANDOMIZED EXPERIMENT TO TARGET HIGH RISK GUN OFFENDERS IN DENVER, COLORADO

Principal Investigator: Andrew Wheeler, PhD

andrew.wheeler@crimede-coder.com

Principal Police Liaison: Captain Jacob Herrera

Jacob.Herrera@denvergov.org

Funding Recipient: [CRIME De-Coder LLC](#)

10/27/2024

Research Question

While hotspots policing has garnered an ample evidence based through many randomized trials, targeting high risk individuals has not. The majority of evaluations of chronic offender initiatives to date have not been randomized experiments and have shown mixed evidence of efficacy. For example, one quasi-experimental predictive policing evaluation in Chicago did not show efficacy (Saunders et al., 2016), and another social service oriented intervention did show some limited evidence of efficacy (Bhatt et al., 2024).

We know a small number of individuals contribute to a large proportion of crime (Spelman 1990; Wood & Papachristos, 2019), and that systems to effectively forecast those individuals based on information known to police is feasible (Bhatt et al., 2024; Wheeler et al., 2019). But despite their popularity in policing, we have very little evidence of the overall efficacy of chronic offender initiatives within police departments.

Building a stronger evidence base for chronic offender initiatives is important for two reasons. One reason is that it is important to establish whether such an intervention is worthy to be considered a reasonable evidence based policy to promote its application to other police departments, as hotspots policing commonly is now. The other reason is that chronic offender initiatives, when focusing on punitive policing measures (such as surveillance or arrests), have a much greater potential for secondary harms to individuals. Knowing if *they don't work* (and thus advocating to avoid them in practice) is just as important as knowing whether they do work.

This research, in collaboration with the Denver Police Department, is interested in constructing a randomized trial to determine the efficacy of a chronic offender initiative targeted towards individuals at high risk of gun violence. Denver PD has already engaged in tactics such as leveraging more detective resources to greatly increase clearances of shootings (Alcorn, 2023), as well as place based interventions (Herold et al., 2020). Denver PD wishes to expand their current use of evidence based practices to include a person based component.

Study Design and Timeline

The researchers, based on Denver's historical crime and arrest data, will develop a predictive model to identify individuals that have a high probability of being involved in future gun violence. The researcher partner has previously been involved with a chronic offender prediction model (Wheeler et al., 2019), as has won several predictive competitions (such as those sponsored by NIJ and NASA).

Among those top-k individuals predicted to be high risk, the researcher will then randomize whom is to be chosen for the particular intervention on some regular interval. This will provide something a true exogenous event to determine a

counterfactual estimate of the future violence (either as victims or offenders) among those targeted for treatment.

The main outcomes of interest will be the cumulative number of violent offenses involving a gun and gun specific crimes (such as illegally carrying a weapon) that the offenders commit, as well as violent victimizations involving a gun.

It is necessary to have a sufficient timeline in this project to *detect* whether the intervention is effective. Appendix A details a power analysis simulation, using Denver PD data, to approximate what a realistic intervention will look like in practice. We plan on analyzing preliminary outcomes at year 1, but expect the intervention to take around 2 years before it is well enough powered to detect a 20% reduction in violent and gun crime outcomes for those treated. We plan on conducting the outcome evaluation for three years, so those whom get the treatment have at least two years of post-intervention data.

Much of the funding will be focused on generating a real time system embedding in Denver PD to identify and track individuals they have targeted for intervention. Once those fixed processes have been put into place, it is merely monitoring the progress of the system over time.

Treatment

In terms of what Denver PD will specifically do with based on the identified high risk individuals, they wanted to conduct a multi-pronged approach. First, and the main treatment, is they will conduct home visits, with an officer and a case-worker visiting the individual and providing potential social work resources. This is similar to a custom notification approach that is sometimes employed in focused deterrence initiatives (Hunt et al., 2017).

The second component was assigning individuals to a longer term team to identify effective approaches to limit their future criminality. Similar to problem oriented policing (Hinkle et al., 2020), this is not scripted what exactly the intervention may look like. It is possible it will look like traditional policing tactics, such as building long term investigations (in the case the individual is involved in other illicit activities), or other ad-hoc interventions which may be more social service like in nature (such as helping the individual get a job).

This then potentially confounds the multiple treatments – home visits, social services, and longer term interventions by Denver PD. To potentially untangle these confounds, the Denver PD crime analysis unit will build case-profiles for offenders. This will include information on whether the offender worked with the social worker, and whether they were further prioritized in any Denver PD operation.

These case profiles will be qualitative, but it may be possible to later identify regular types of interventions and conduct sub-analysis on those types of

interventions (or conduct treatment-on-treated analysis for taking up social work services). This is similar to the mediation analysis Braga and Bond (2008) conducted in reference to a hotspot problem oriented policing approach.

Written Deliverables

For an approximate timeline and deliverables:

- Months 1-3, predictive model plus automated code to identify high risk, pre-analysis plan for outcome evaluations
- Months 3-9, initial start of the program
- Month 6-9, automation of dashboards and reports to conduct continuous monitoring over time, monitoring of outcomes
- Months 12-18, Preliminary outcome report to Arnold (after intervention has been run for 12 months), initial conference presentations
- Month 24-40, Full outcome report to Arnold for final results (after intervention has been run for 36 months), conference presentations of final results, publication pre-print + journal submission.

As stated, it may be necessary to run the experiment for 2 years to reduce the standard error of the estimate to sufficient precision to detect a 20% reduction. The project team after year 1 will collaborate with Arnold on next steps and expected further deliverables, although it is anticipated almost all analysis and reporting will be automated at that stage.

For an estimated breakdown of the cost, at \$350 per hour and 270 total hours.

- Developing the predictive model and pre-registration, 40 hours
- Automating identification of chronic offenders, 40 hours
- Building a dashboard for continuous monitoring, 40 hours
- Scholarly outputs (peer reviewed paper, conference presentations), 60 hours
- Presentation and on-sight feedback, 100 hours

We also would like to plan for site visits to Denver at approximately months 6 and 18, and 30 to meet with the team and deliver the preliminary results. As well as a budget to present the results at the *American Society of Evidence Based Policing* conference in 2026. Thus we are adding an additional \$9,800 to the budget for travel, room, board, and conference fees (for travel for both Dr. Wheeler and Capt. Herrera). This brings the total estimated budget to *\$107,800*.

In addition to presenting the work in conferences and peer reviewed papers, we also commit to open source all of the work derivative. Including, but not limited to:

- Machine learning algorithms developed to identify chronic offenders
- Any automated computer code used for creating dashboards
- Replication materials to show how any treatment effect estimates resulting from the intervention were calculated

Code and data (in cases where private data can be shared) will be posted in open-source Github repositories.

Study Team

The researchers involved have not previously received funding from Arnold Ventures, although each has ample experience conducting crime reduction initiatives and outcome evaluations in either as a researcher or directly as a police officer.

Principal Investigator Andrew Wheeler: Dr. Andrew Wheeler has a PhD in criminal justice from SUNY Albany, and has collaborated with police departments across the United States. His research has focused on outcome analyses, predictive policing applications, and operations research. He is currently a data scientist in the private sector, and has a consulting firm, CRIME De-Coder, in which he conducts research with police departments. His CV can be viewed at <https://andrewpwheeler.com/c-v/>.

Principal Police Liason, Captain Jacob Herrera of the Denver Police Department: Jacob Herrera is a 17-year veteran of the Denver Police Department. He is currently a Commander in the Chief's office where he is overseeing a comprehensive gun crime strategy. He was a winner of the 2023 IACP 40 under 40 award, and is committed to building the evidence base of police interventions.

References

- Alcorn, T. (2023). *One City's Surprising Tactic to Reduce Gun Violence: Solving More Nonfatal Shootings*.
<https://www.themarshallproject.org/2023/10/30/nonfatal-shootings-police-clearance-rates-denver>.
- Braga, A. A., & Bond, B. J. (2008). Policing crime and disorder hot spots: A randomized controlled trial. *Criminology*, 46(3), 577-607.
- Bhatt, M. P., Heller, S. B., Kapustin, M., Bertrand, M., & Blattman, C. (2024). Predicting and preventing gun violence: An experimental evaluation of READI Chicago. *The Quarterly Journal of Economics*, 139(1), 1-56.
- Herold, T. D., Engel, R. S., Corsaro, N., & Clouse, S. L. (2020). *Place network investigations in Las Vegas, Nevada: Program review and process evaluation*.
- Hinkle, J. C., Weisburd, D., Telep, C. W., & Petersen, K. (2020). Problem-oriented policing for reducing crime and disorder: An updated systematic review and meta-analysis. *Campbell Systematic Reviews*, 16(2), e1089.

- Hunt, P., Parast, L., & Weinberger, G. (2017). *Can an informative letter reduce gun crime and be cost-effective? A study of Los Angeles*. Rand, Santa Monica, CA. https://www.rand.org/pubs/research_reports/RR1934.html
- Saunders, J., Hunt, P., & Hollywood, J. S. (2016). Predictions put into practice: a quasi-experimental evaluation of Chicago's predictive policing pilot. *Journal of Experimental Criminology*, 12, 347-371.
- Spelman, W. (1990). *Repeat Offender Programs for Law Enforcement*. Police Executive Research Forum. Washington, D.C.
- Wheeler, A. P., Worden, R. E., & Silver, J. R. (2019). The accuracy of the violent offender identification directive tool to predict future gun violence. *Criminal Justice and Behavior*, 46(5), 770-788.
- Wheeler, A.P. (2024a). *Poisson designs and minimal detectable effect sizes*. <https://andrewpwheeler.com/2024/03/18/poisson-designs-and-minimum-detectable-effects/>
- Wheeler, A.P. (2024b). *CompStat and Counterfactuals*. <https://crimede-coder.com/blogposts/2024/CompstatCounter>.
- Wood, G., & Papachristos, A. V. (2019). Reducing gunshot victimization in high-risk social networks through direct and spillover effects. *Nature Human Behaviour*, 3(11), 1164-1170.